

- `analogWrite()` vs `analogRead()`:

- `analogWrite(pin, value)`:

- Writes a PWM signal to a pin.
- Value range: 0 to 255 (0 = OFF, 255 = fully ON).
- Used to control devices like LED brightness or motor speed.
- IoI Example: Controlling brightness of an LED remotely using a mobile app.

- `analogRead(pin)`:

- Reads an analog voltage from a pin.
- Value range: 0 to 1023 (0V to 5V).
- Used to read data from analog sensors like temperature or light sensors.
- IoI Example: Reading temperature from a thermistor and sending it to the cloud.

- PWM (Pulse Width Modulation):

- PWM is a technique where the width of the HIGH pulse is varied to simulate an analog output.
- It allows us to control power delivered to devices like LEDs or motors.
- Used because Arduino pins can only output HIGH (5V) or LOW (0V), not real analog voltages.
- IoI Example: Dimming a LED in a smart lighting system using PWM.